

ORIGINAL ARTICLE

The functional effects of BIO-Enriched protein flours on biochemical outcomes and visceral adipose tissue in a ketogenic diet

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ABSTRACT

Background and aim: This study examined the effects of a ketogenic diet enriched with organic (BIO) protein-fortified flours supplemented with essential amino acids and B vitamins.

Methods: The research was conducted at the SANABEA Agricultural Company, comparing industrial and BIO flour production, followed by a controlled dietary intervention. Twenty participants (10 males and 10 females, aged 30–40 years, BMI 35–39.9 kg/m²) underwent a three-phase nutritional protocol: Very Low-Calorie Ketogenic Diet (VLCKD), Low-Calorie Diet (LC), and a Nutritional and Weight Maintenance phase (EA). Subjects were divided into three groups based on the source of dietary protein. Anthropometric and biochemical parameters were assessed, including body weight, BMI, visceral adipose tissue (VAT), Orexin-A, and Adiponectin levels.

Results: The intervention resulted in significant reductions in body weight, BMI, fat mass, and VAT, alongside increased levels of Orexin-A and Adiponectin. The ketogenic diet demonstrated beneficial effects on metabolic and inflammatory markers, notably reducing C-reactive protein levels and enhancing anti-inflammatory responses. The



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inclusion of BIO protein-enriched flours helped mitigate potential nutritional im-balances commonly associated with ketogenic diets.

Conclusions: These findings underscore the im-portance of a multidimensional nutritional approach that considers metabolic, inflammatory, and gut microbiota interactions. Future studies should expand the sample size and further investigate the longterm effects of BIO-enriched flours in dietary interventions.

Key words: ketogenic diet (KD), BIO-enriched protein flours; metabolic health, visceral adipose tissue, inflammation and immune response, Orexin-A and adiponectin

Introduction

In recent years, there has been a growing awareness of the impact that dietary patterns have on human health, quality of life, and the sustainability of food systems. This shift has led to a significant emphasis on nutritional strategies that support not only individual well-being but also environmental stewardship. Consumers are increasingly prioritizing food products that are both high in nutritional value and produced through eco-friendly practices (1). As a result, sustainable agriculture—particularly organic and regenerative farming systems—has gained prominence. These practices aim to preserve biodiversity, improve soil health, and reduce the environmental footprint of food production, while simultaneously enhancing the nutritional profile of agricultural outputs. Within this evolving landscape, cereals have emerged as staple crops of critical importance, both nutritionally and economically. Their widespread consumption, especially in industrialized nations, makes them a key target for nutritional enhancement and innovation. Cereals represent the foundation of the Mediterranean diet, which is globally recognized for its protective effects against chronic diseases and its contribution to longevity. Their role as primary sources of complex carbohydrates, dietary fiber, and plant-based proteins makes them indispensable in both traditional and modern diets (2–4). Beyond their carbohydrate content, cereals contain a moderate but significant proportion of proteins (typically 10–15% of dry weight). These proteins are composed of various fractions, notably albumins and globulins, which are rich in essential amino acids such as lysine, leucine, glutamine, and arginine critical for muscle synthesis, immune function,

and cellular repair. Additionally, whole grains provide a rich matrix of micronutrients and phytochemicals, including phenolic acids, tocopherols (vitamin E), carotenoids, and flavonoids. These compounds exhibit potent antioxidant activity, helping to neutralize free radicals, reduce oxidative stress, and modulate inflammatory responses (5–7). Whole cereals are also significant sources of lignans and phytosterols, compounds associated with cardiovascular protection and lipid regulation. Furthermore, they contribute substantially to the intake of folates and other B-complex vitamins, essential for DNA synthesis, red blood cell formation, and neurological function. The high dietary fiber content in whole grains supports gastrointestinal health, promotes satiety, and assists in weight management. Collectively, these components work synergistically to reduce the risk of non-communicable diseases such as type 2 diabetes, obesity, atherosclerosis, and certain cancers (8). At the same time, scientific attention has increasingly focused on the relationship between visceral adiposity and systemic inflammation. Visceral fat accumulation is now understood as a key driver of chronic low-grade inflammation, a pathophysiological condition characterized by altered secretion of adipokines and cytokines. Enlarged adipocytes in visceral fat depots promote the production of pro-inflammatory molecules such as TNF- α and IL-1 β , while downregulating beneficial adipokines like adiponectin (9). This imbalance contributes to metabolic dysfunctions including insulin resistance, dyslipidemia, and endothelial damage, thereby increasing the risk for metabolic syndrome and cardiovascular disease (10). In this context, the ketogenic diet (KD) has emerged as a promising nutritional intervention. By inducing a state of physiological ketosis where ketone bodies such

as β -hydroxybutyrate become the primary energy substrate the KD promotes fat oxidation, reduces insulin levels, and lowers inflammation. Ketosis is associated with enhanced adiponectin levels, suppressed production of inflammatory cytokines, and modulation of hypothalamic pathways involved in appetite regulation. The anti-inflammatory and antioxidative properties of ketone bodies also support improved cellular metabolism and mitochondrial function, making KD an effective strategy for managing obesity, metabolic syndrome, and related inflammatory conditions (8–10). Given these health-promoting effects, there is growing interest in optimizing the ketogenic diet to ensure both efficacy and nutritional balance. Traditional KD regimens, often high in animal-based fats and low in micronutrients, may pose risks of nutritional deficiencies or poor adherence (11,12). To address these limitations, the present study explores the functional impact of incorporating BIO-enriched protein flours into a structured ketogenic dietary protocol. These flours are derived from whole grains that have been selectively processed to reduce carbohydrate content while enriching protein fractions and fortifying essential micronutrients such as methionine, cysteine, and B vitamins. The use of these functional flours in common food formulations—such as pasta, bread, and biscuits—offers an innovative and accessible means to align ketogenic principles with organic, plant-based nutrition. This integrative approach seeks not only to preserve the metabolic and anti-inflammatory benefits of ketosis but also to elevate the overall nutritional value of the diet. By bridging the gap between sustainable agriculture and therapeutic nutrition, this study aims to demonstrate how BIO-enriched cereal derivatives can contribute to healthier dietary practices and a reduction in chronic disease burden through improved metabolic and inflammatory outcomes.

Materials and methods

Production of flours and products for the ketogenic diet

At the SANABEA Agricultural Company, all phases of flour production were analyzed to determine the differences between industrial and organic

(BIO) production. A quality control assessment was conducted, analyzing the protein content in terms of essential amino acids and B vitamins in the enriched flours. Additionally, products suitable for inclusion in the ketogenic dietary regimen were selected, and subsequent dietary plans were designed for administration to the study group. The aim was to evaluate the beneficial effects of a ketogenic diet and to compare the BIO dietary regimen with conventional food-based diets.

Participants

A total of 20 subjects aged 30–40 years (10 males and 10 females) with a BMI of 35–39.9 kg/m² were enrolled. They underwent dietary therapy consisting of three distinct phases: the “Ketogenic Diet” (VLCKD), the “Low-Calorie Diet” (LC), and the “Nutritional and Weight Balance” (EA), conducted at the Department of Experimental Medicine, Section of Human Physiology, Unit of Dietetics and Sports Medicine, University of Campania “Luigi Vanvitelli”. The VLCKD phase involved a daily carbohydrate intake of <50 g and a maximum caloric intake of 800 kcal/day, distributed as follows: carbohydrates (25%), lipids (35%), and proteins (40%). Participants were randomly assigned to three groups: A group consuming exclusively BIO protein-enriched flours with a protein content of approximately 20%, enriched with essential amino acids such as methionine and cysteine. A second group following a ketogenic diet in which protein intake was derived equally from both animal and plant sources originating from organic farming and livestock. A third group following a ketogenic diet where protein intake was exclusively derived from animal sources obtained from organic livestock. All enrolled participants were in good health according to the ESH/ESC 2007 guidelines. They provided written informed consent, and the study was approved by the local Ethics Committee (AOU Università della Campania Vanvitelli - Prot. 0003232/i del 01/02/2023). The VLCKD phase was maintained until participants achieved a 20% weight loss over a period of 3 months. The second phase (LC) involved the reintroduction of carbohydrates, with a daily caloric intake of approximately 1000–1200 kcal/day, distributed as 55% carbohydrates, 25% fats, and 20% proteins, lasting about 3 months. Subsequently, participants followed an

EA program for 6 months, maintaining a normocaloric intake distributed as 55% carbohydrates, 30% fats, and 15% proteins.

Anthropometric and biochemical assessments

Anthropometric and biochemical parameters were assessed in all study participants at baseline (T0) and at the end of the intervention (T1):

- Body weight, height, body mass index (BMI), and waist circumference (WC).
- Body composition analysis using bioelectrical impedance analysis (BIA Akern, IT) and visceral adipose tissue (VAT) assessment using DXA (GE Healthcare, USA).
- Evaluation of key biochemical parameters, reactive oxygen species (ROS) levels via the d-ROMs test, and serum levels of Orexin-A and Adiponectin.

Statistical analysis

All statistical analyses were performed using GraphPad Prism version 8.0.2 (GraphPad Software, San Diego, CA, USA). Data are presented as mean $\pm$ standard deviation (SD). The normality of data distribution was assessed using the Shapiro-Wilk test. For normally distributed variables, paired t-tests were conducted to compare baseline (T0) and follow-up (T1) values. In cases where data did not follow a normal distribution, the Wilcoxon signed-rank test was applied. A significance level of $p < 0.05$ was considered statistically significant. Furthermore, linear regression analyses were conducted to evaluate the relationships between changes (Δ) in Orexin-A, visceral adipose tissue (VAT), and C-reactive protein (CRP) levels. The strength of these associations was assessed by calculating the coefficient of determination (R^2), with statistical significance set at $p < 0.05$.

Results

The analysis of metabolic and inflammatory parameters in obese subjects undergoing a ketogenic diet with organic products revealed significant

improvements from baseline (T0) to follow-up (T1). Notably, body weight decreased significantly from 92.23 ± 17.11 kg at T0 to 79.43 ± 11.36 kg at T1 ($p < 0.001$), leading to a significant reduction in BMI from 33.19 ± 4.78 kg/m² to 27.86 ± 3.62 kg/m² ($p < 0.001$). Lipid profile improvements were also observed, with total cholesterol decreasing significantly from 221.11 ± 50.77 mg/dl to 174.91 ± 32.93 mg/dl ($p < 0.05$), and LDL cholesterol levels showing a marked reduction from 141.83 ± 36.48 mg/dl to 107.57 ± 27.72 mg/dl ($p < 0.05$). Triglycerides also decreased significantly (137.40 ± 125.27 mg/dl to 83.35 ± 25.14 mg/dl, $p < 0.05$), while HDL cholesterol levels did not change significantly. Inflammatory markers and liver enzymes exhibited notable changes, with CRP levels significantly decreasing from 0.89 ± 0.1 mg/ml to 0.48 ± 0.07 mg/ml ($p < 0.05$), suggesting a reduction in systemic inflammation. Liver enzyme levels, including AST (21.27 ± 5.98 U/L to 23.31 ± 11.47 U/L, $p < 0.05$) and ALT (26.51 ± 14.89 U/L to 26.06 ± 16.27 U/L, $p < 0.05$), showed slight but significant variations. Gamma GT levels demonstrated a significant decrease from 31.19 ± 19.88 U/L to 15.31 ± 5.41 U/L ($p < 0.05$), further indicating improved liver function. Regarding hormonal markers, a substantial increase in Orexin-A levels was observed, rising from 9.91 ± 0.27 pg/ml to 16.24 ± 1.33 pg/ml ($p < 0.001$). Adiponectin levels also increased significantly from 11.2 ± 1.7 μ g/ml to 24.33 ± 0.37 μ g/ml ($p < 0.001$), suggesting potential improvements in metabolic regulation (Table 1).

Body composition analysis revealed significant reductions in visceral adipose tissue (VAT) and fat mass following the ketogenic diet. VAT decreased from 1541.55 ± 141.63 g at T0 to 927.79 ± 104.92 g at T1 ($p < 0.001$), while fat mass significantly decreased from 39208.77 ± 1432.55 g to 27377.0 ± 1217.48 g ($p < 0.001$).

Lean mass, however, remained relatively stable, with no significant change observed (48789.57 ± 1712.36 g at T0 vs. 48093.68 ± 1670.65 g at T1, ns), indicating the diet's effectiveness in preserving muscle mass. Bone mineral density also showed no significant variation (1225.57 ± 21.23 vs. 1229.31 ± 21.46 , ns), suggesting the dietary intervention did not negatively impact bone health (Table 2).

A linear regression analysis was conducted to investigate the relationships between changes (Δ) in

Table 1. Changes in metabolic and inflammatory parameters in obese subjects following a ketogenic diet with organic products.

	Obese subjects undergoing a ketogenic diet with organic products		p-value
	T0	T1	
Age	38 ± 6.4		ns
Height (m)	1.67 ± 0.11		ns
Weight (kg)	92.23 ± 17.11	79.43 ± 11.36	<0.001
BMI (kg/m ²)	33.19 ± 4.78	27.86 ± 3.62	<0.001
Total cholesterol (mg/dl)	221.11 ± 50.77	174.91 ± 32.93	<0.05
HDL (mg/dl)	55.13 ± 11.14	47.76 ± 9.14	ns
LDL (mg/dl)	141.83 ± 36.48	107.57 ± 27.72	<0.05
Triglycerides (mg/dl)	137.40 ± 125.27	83.35 ± 25.14	<0.05
Insulinemia (μU/ml)	10.53 ± 7.18	5.37 ± 3.79	<0.05
Total proteins (g/dl)	7.30 ± 0.4	7.13 ± 0.4	ns
AST-GOT (U/L)	21.27 ± 5.98	23.31 ± 11.47	<0.05
ALT-GPT (U/L)	26.51 ± 14.89	26.06 ± 16.27	<0.05
Gamma GT (U/L)	31.19 ± 19.88	15.31 ± 5.41	<0.05
Blood urea nitrogen (mg/dl)	35.35 ± 8.43	34.68 ± 9.16	ns
CRP (mg/ml)	0.89 ± 0.1	0.48 ± 0.07	<0.05
Orexin-A (pg/ml)	9.91 ± 0.27	16.24 ± 1.33	<0.001
Adiponectin (μg/ml)	11.2 ± 1.7	24.33 ± 0.37	<0.001

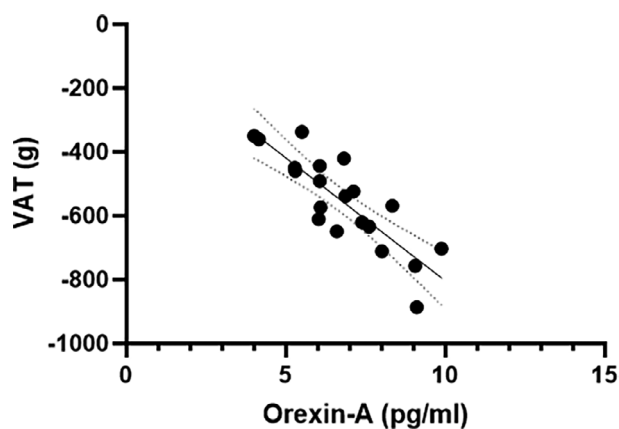
BMI = Body Mass Index; HDL = High-Density Lipoprotein; LDL = Low-Density Lipoprotein; CRP = C-Reactive Protein; AST = Aspartate Aminotransferase; ALT = Alanine Aminotransferase; Gamma GT = Gamma-Glutamyl Transferase.

Orexin-A, visceral adipose tissue (VAT), and C-reactive protein (CRP) levels between baseline (T0) and follow-up (T1). The analysis revealed a strong positive association between Δ Orexin-A and Δ VAT, with an R^2 value of 0.70, indicating that 70% of the variance in VAT changes can be explained by changes in

Table 2. Changes in body composition parameters in obese subjects following a ketogenic diet with organic products.

	Obese subjects undergoing a ketogenic diet with organic products		p-value
	T0	T1	
Visceral Adipose Tissue (g)	1541.55 ± 141.63	927.79 ± 104.92	<0.001
Fat Mass (g)	39208.77 ± 1432.55	27377.0 ± 1217.48	<0.001
Lean Mass (g)	48789.57 ± 1712.36	48093.68 ± 1670.65	ns
Bone Mineral Density	1225.57 ± 21.23	1229.31 ± 21.46	ns

VAT = Visceral Adipose Tissue; BMI = Body Mass Index.

**Figure 1.** Linear regression analysis between changes in Orexin-A (Δ Orexin-A) and visceral adipose tissue (Δ VAT) levels from baseline (T0) to follow-up (T1).

VAT = Visceral Adipose Tissue, Δ = Change between T0/T1.

Orexin-A levels ($F = 43.34$, $p < 0.001$). This suggests a significant influence of Orexin-A on VAT accumulation over time (Figure 1).

Additionally, a moderate relationship was found between Δ Orexin-A and Δ CRP, with an R^2 value of 0.50 ($F = 18.53$, $p < 0.001$). These results suggest that fluctuations in Orexin-A levels contribute to inflammatory marker variations, supporting a potential link between Orexin-A and systemic inflammation (Figure 2).

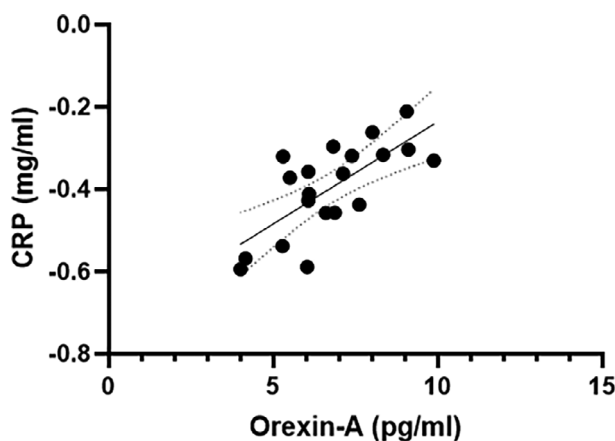


Figure 2. Linear regression analysis between changes in Orexin-A (Δ Orexin-A) and C-reactive protein (Δ CRP) levels from baseline (T0) to follow-up (T1).

CRP = C-Reactive Protein, Δ = Change between T0/T1.

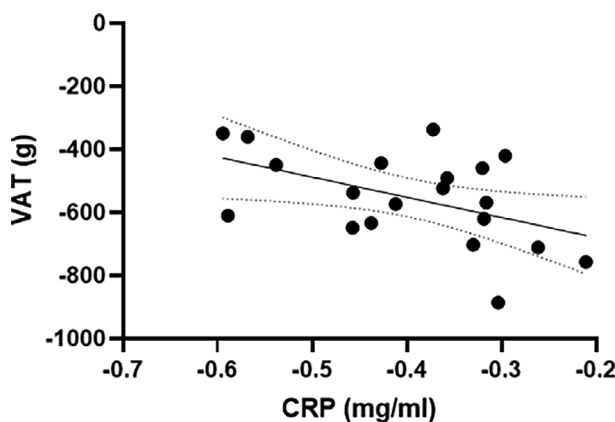


Figure 3. Linear regression analysis between changes in visceral adipose tissue (Δ VAT) and C-reactive protein (Δ CRP) levels from baseline (T0) to follow-up (T1).

VAT = Visceral Adipose Tissue, CRP = C-Reactive Protein, Δ = Change between T0/T1.

Finally, a weaker but statistically significant correlation was observed between Δ VAT and Δ CRP, with an R^2 value of 0.23 ($F = 5.57$, $p < 0.05$), suggesting that changes in visceral adiposity may partially contribute to variations in inflammation markers (Figure 3).

Overall, these findings demonstrate significant relationships between the analyzed metabolic and inflammatory parameters, highlighting the potential regulatory role of Orexin-A in adipose tissue dynamics and inflammation.

Discussion

The results of this study strongly support the efficacy of a ketogenic dietary protocol enriched with organic (BIO) protein flours fortified with essential amino acids and B vitamins in improving both anthropometric and biochemical parameters. The intervention led to significant reductions in body weight, body mass index (BMI), fat mass, and visceral adipose tissue (VAT). Concurrently, an increase in circulating levels of Orexin-A and Adiponectin was observed, indicating improved neuroendocrine and metabolic regulation. These findings align with previous studies that underscore the metabolic and anti-inflammatory benefits of ketogenic diets, especially when carefully formulated to avoid micronutrient deficiencies (13,14). One of the most notable outcomes was the strong correlation between weight loss and reduction in VAT—a form of adiposity widely associated with cardiometabolic risk. VAT plays a central role in the pathophysiology of metabolic diseases by promoting chronic low-grade inflammation, primarily through the dysregulated secretion of adipokines and pro-inflammatory cytokines such as $\text{TNF-}\alpha$ and $\text{IL-1}\beta$. This inflammatory cascade has been shown to disrupt endocrine and immune system functions, particularly by impairing lymphatic tissue and altering immune cell populations within adipose depots (15). In this context, the decrease in serum C-reactive protein levels observed in our cohort reinforces the anti-inflammatory potential of the ketogenic intervention, consistent with existing evidence linking VAT reduction to decreased systemic inflammation (16). The dietary intervention implemented in this study followed the principles of the very low-calorie ketogenic diet (VLCKD), which induces ketosis through significant restriction of carbohydrates and total energy intake. This metabolic shift promotes hepatic ketogenesis, leading to increased fat oxidation, glycogen depletion, and reduced hepatic glucose output. As fat depots in the liver and muscle tissues diminish, insulin resistance is alleviated, improving glucose homeostasis and restoring pancreatic β -cell function (17). Additionally, ketosis has been shown to exert central effects on appetite control by increasing satiety hormones and modulating hypothalamic pathways. Our findings support this, as improved satiety

and better dietary adherence were commonly reported by participants during the intervention. Interestingly, the ketogenic diet's impact extends beyond metabolic and endocrine systems to include effects on oxidative stress and neurological health. The increase in Orexin-A levels observed may reflect improved hypothalamic function, as orexins play a crucial role in regulating arousal, energy balance, and appetite. Ketone bodies have been shown to modulate neuronal energy metabolism by enhancing mitochondrial efficiency and reducing reactive oxygen species (ROS) production. Our data confirm this, with d-ROMs assays indicating reduced oxidative stress at the hepatic and hypothalamic levels. These changes are relevant, as hypothalamic ROS signaling is a key determinant of energy intake regulation through the NADPH oxidase pathway. By limiting ROS accumulation, ketosis may promote satiety and reduce appetite-driven caloric intake (18). The inclusion of BIO-enriched protein flours in the ketogenic dietary protocol represents a significant innovation. These flours not only provide essential amino acids—such as methionine and cysteine—but also restore key micronutrients like B vitamins that are often insufficient in standard ketogenic regimens. This nutritional enrichment helps counteract the potential deficiencies associated with restrictive diets, improving overall diet quality and functional outcomes (19). Moreover, the formulation of commonly consumed foods (e.g., pasta, bread, biscuits) using these flours increases dietary acceptability and sustainability, which are crucial for long-term compliance and success. From a translational standpoint, this research bridges nutritional science with sustainable food production. The organic cultivation and processing of enriched flours contribute to environmental stewardship and offer products that are not only health-promoting but also eco-friendly. Furthermore, their consumption may positively influence gut microbiota composition—a key determinant of metabolic health. The gut microbiota plays an active role in energy regulation, satiety, and nutrient absorption via the gut-brain axis. Emerging evidence suggests that ketogenic diets can shift microbiota composition toward a more anti-inflammatory profile. When combined with organic, fiber-containing ingredients, these changes may be further enhanced, contributing to improved intestinal barrier

function and systemic metabolic balance (20–22). Notably, the interaction between gut-derived metabolites such as butyrate and circulating ketone bodies may yield synergistic effects, promoting both intestinal health and systemic ketone utilization. Studies by Crawford et al. have shown that myocardial tissue, in the context of a well-regulated gut microbiota, becomes more efficient at oxidizing ketone bodies for energy production (23). These findings point to a systemic benefit of nutritional strategies that integrate gut microbiota modulation with metabolic interventions like the ketogenic diet. Our study supports this integrative model and encourages further exploration into how diet-driven shifts in microbiota can optimize clinical outcomes in obese and metabolically impaired populations (24–28). This investigation also highlights the value of a multidisciplinary and participatory approach, linking scientific research with agricultural innovation and food technology (24,29,30). The study benefited from direct collaboration with production facilities, enabling the real-world application of laboratory findings. The development and implementation of BIO-enriched protein flours not only enhanced the nutritional profile of the ketogenic diet but also contributed to the creation of practical, market-ready solutions with potential for broader public health impact. While the results are promising, the study has some limitations—most notably the relatively small sample size and limited duration. These constraints call for expanded trials with larger populations and longer follow-up periods to validate and extend the observed benefits. Future research should also explore the long-term effects of ketogenic diets using BIO-enriched flours on gut microbiota diversity, cognitive function, and disease prevention in at-risk populations.

Conclusions

This study demonstrates that integrating BIO protein-enriched flours fortified with essential amino acids and B vitamins into a ketogenic dietary protocol can significantly enhance both anthropometric and biochemical outcomes. This nutritional strategy not only supports effective weight management and metabolic improvement but also aligns with sustainable

agriculture and functional food development. The findings lay a foundation for further investigations aimed at refining ketogenic interventions through bio-functional, eco-conscious dietary innovations.

Author Contributions: Antonietta Messina and Antonietta Monda contributed equally to this work. Giovanni Messina and Marcellino Monda conceived and supervised the study. Fiorenzo Moscatelli, Marco La Marra, and Emanuele Capasso were responsible for data collection and experimental procedures. Vincenzo Monda, Paride Vasco, and Mario Chisari contributed to data analysis and interpretation. Lucio Di Mauro, Girolamo Di Maio, and Ines Villano performed the statistical analysis and assisted in manuscript preparation. Maria Casillo and Rita Polito contributed to the literature review and manuscript editing. All authors read and approved the final version of the manuscript.

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